King County Metro Knowledge Graph

Transforming Static GTFS Data into Dynamic Visualizations

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1. Information Story
2. What Existed
3. What We Did
4. What Exists Now

Existing Information

Current GTFS File

```
agency_id,route_id,route_short_name,route_long_name,route_type,route_desc,route_url,rout-
e_color,route_text_color,route_sort_order
1,100001,1,,3,Kinnear - Downtown Seattle,https://kingcounty.gov/en/dept/metro/routes-and-
service/schedules-and-maps/001.html,FDB71A,000000,
1,100002,10,,3,Capitol Hill - Downtown Seattle,https://kingcounty.gov/en/dept/metro/
routes-and-service/schedules-and-maps/010.html,FDB71A,000000,
1,100003,101,,3,Renton Transit Center - Downtown Seattle,https://kingcounty.gov/en/dept/
metro/routes-and-service/schedules-and-maps/101.html,FDB71A,000000,
1,100004,105,,3,Renton Highlands - Renton Transit Center,https://kingcounty.gov/en/dept/
metro/routes-and-service/schedules-and-maps/105.html,FDB71A,000000,
1,100005,106,,3,Renton Transit Center - Skyway - Downtown Seattle,https://kingcounty.gov/
en/dept/metro/routes-and-service/schedules-and-maps/106.html,FDB71A,000000,
1,100006,107,,3,Renton Transit Center - Rainier Beach,https://kingcounty.gov/en/dept/
metro/routes-and-service/schedules-and-maps/107.html,FDB71A,000000,
```

Map PDFs

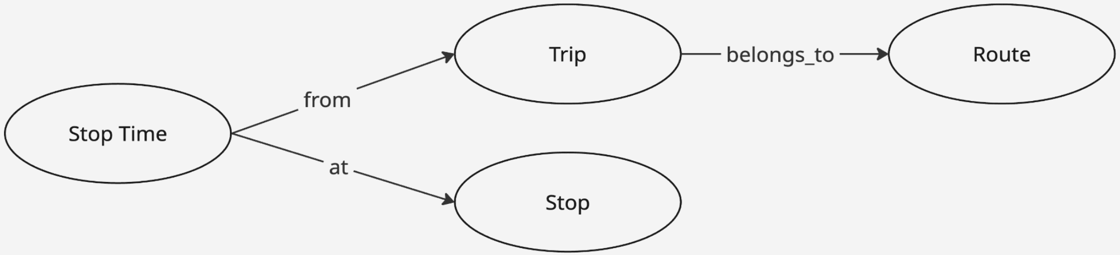
routes.txt ¶

File: **Required**

Primary key (`route_id`)

Field Name	Type	Presence	Description
`route_id`	Unique ID	**Required**	Identifies a route.
`agency_id`	Foreign ID referencing `agency.agency_id`	**Conditionally Required**	Agency for the specified route. Conditionally Required: - **Required** if multiple agencies are defined in agency.txt. - Recommended otherwise.
`route_short_name`	Text	**Conditionally Required**	Short name of a route. Often a short, abstract identifier (e.g may be defined. Conditionally Required: - **Required** if `routes.route_long_name` is empty. - Recommended if there is a brief service designation. This characters.
`route_long_name`	Text	**Conditionally Required**	Full name of a route. This name is generally more descriptiv and `route_long_name` may be defined.

Ontology & Metadata



We kept the relationships and key metadata from the GTFS schema, just reduced in scope

Type	Field	Description
route	route_short_name	route number ex. "3"
	route_description	ex. "Capitol Hill - Downtown Seattle"
stop	stop_name	ex. "1st Ave & Spring St"
	stop_lat	latitude
	stop_lon	longitude
trip	route	route node belonging to trip
	trip_headsign	readable identifier ex. "1st Ave & Spring St"
stop_time	trip	trip node belonging to stop time
	stop	stop node belonging to stop time
	stop_sequence	relative position of stop in trip sequence
	arrival_time	when bus arrives
	departure_time	when bus departs

Transformation of GTFS Format

1

GTFS Files

Zipped .txt files, readable as CSVs

2

Data Conversion

Reformatted CSV rows into RDF templates

3

Construction

Constructed RDF knowledge graph in memory

4

Serialization

Write knowledge graph into N-Triples format to share and import

5

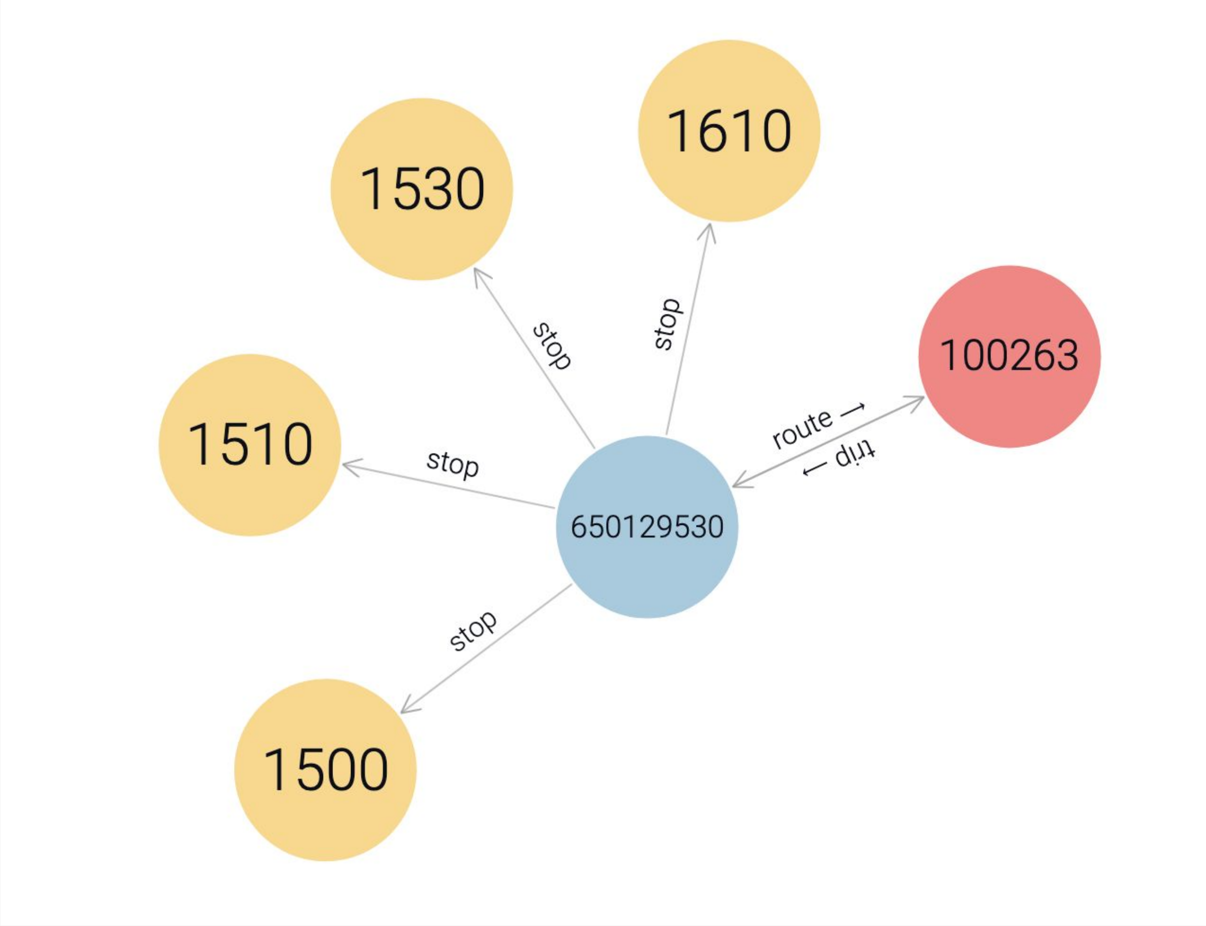
Deployment

Import N-Triples file into graph database, or in memory

Knowledge Graph: The core information structure

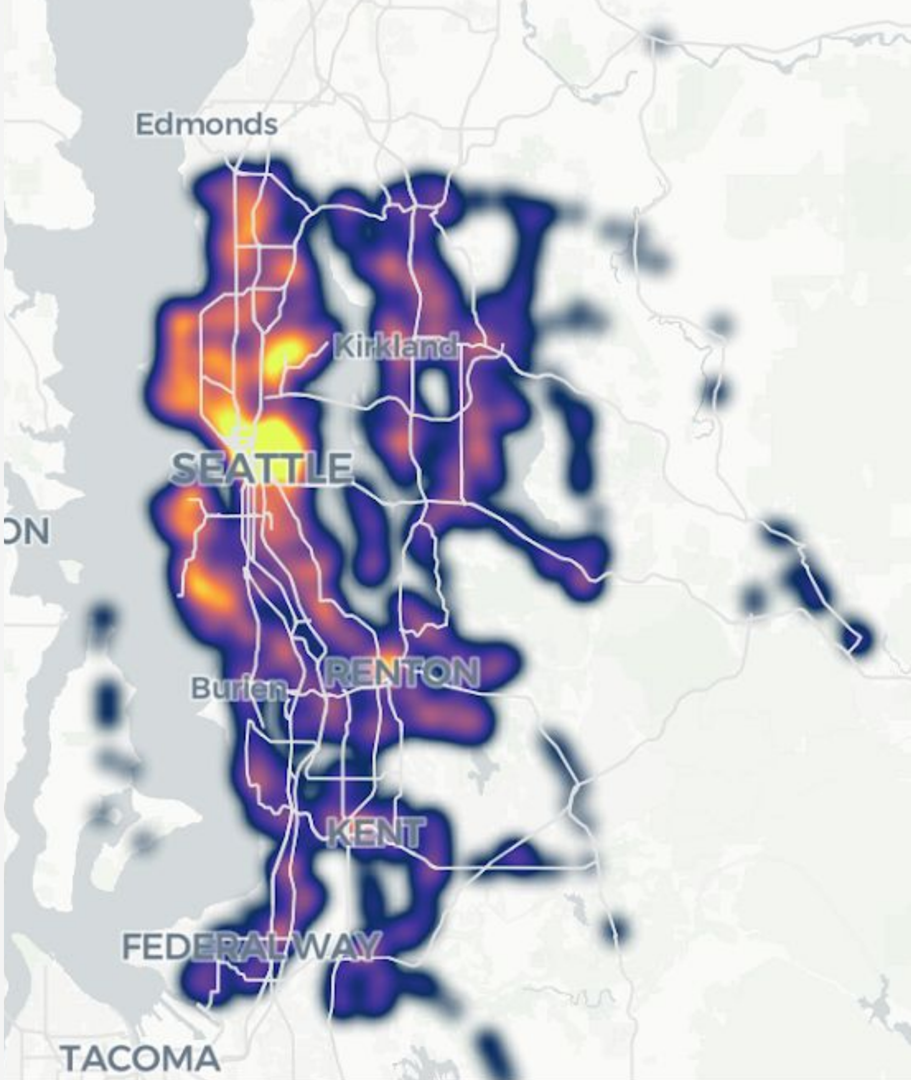
An RDF (Resource Description Framework) following Semantic Web Standards

Queryable via SPARQL endpoint



Putting the Knowledge Graph to Work

Tasks like network analyses highlight the strengths of a graph data structure



Heat map of stops

FAIR Assessment

1

Findable

Our serialization retained the robust findability features of the original GTFS files: schema and metadata remain in place

2

Accessible

N-triple file format is similar to JSON for human readability, while being efficiently machine ingestible

3

Interoperable

N-triples are able to be ingested and displayed by a diversity of programs and databases, allowing for a wide variety of end user applications

4

Reusable

N-triple formats does not lose anything from the original GTFS files, and can be converted back to the original .txt files, or converted to another file type with relative ease

Test Plan

Transit Lab Workflow

- Manually test github installation instructions
- Manually test Jupyter notebook (cell by cell or end to end)
- Check validity of output file

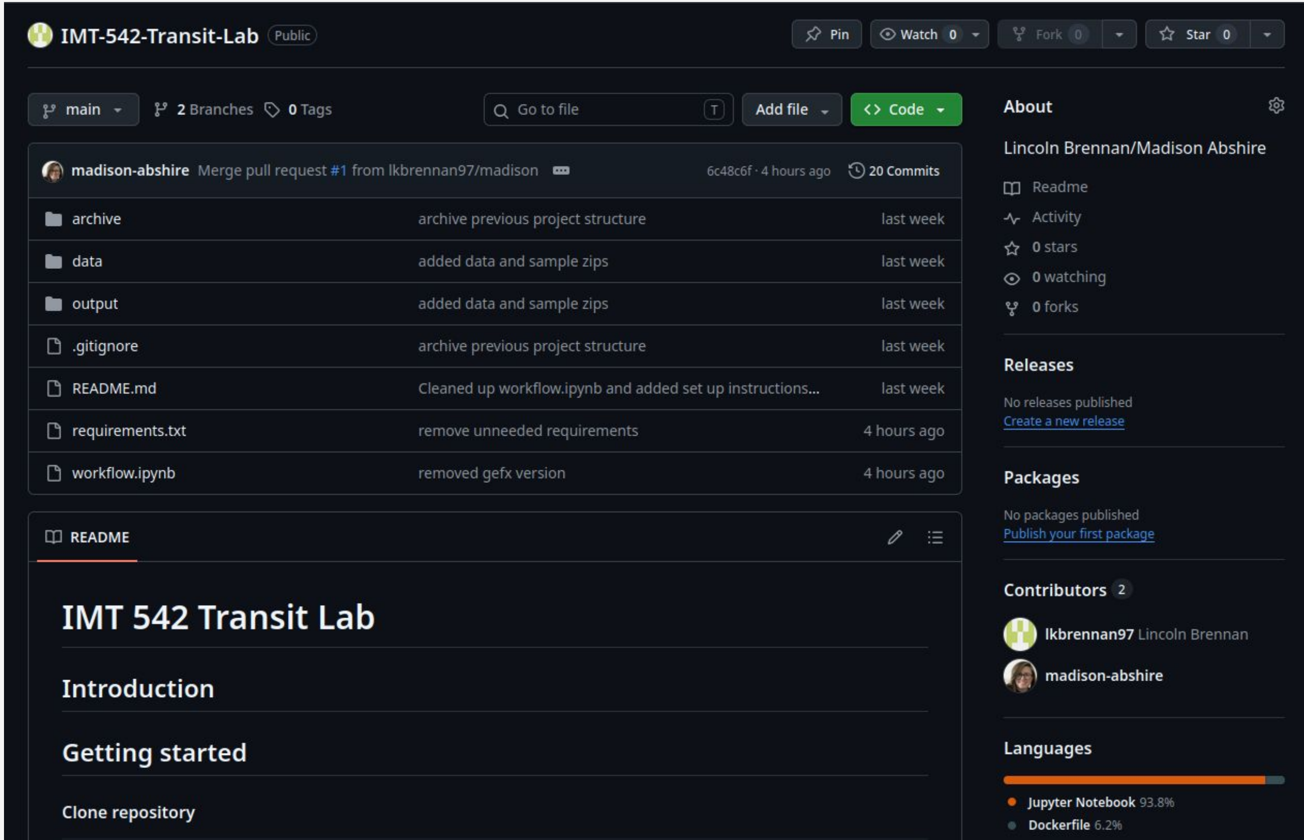
Transit Lab Visualizer

- Unit test suite
- End to end tests from Streamlit front end to GraphDB back end
- Tracking initial query and start up times
- Queries can be tested in the GraphDB Workbench

Performance and Availability

Performance Issues:

- **Time to Serialize**
- **Time to Query**
- **Size of queries**



Ask us anything!